

Spatiotemporal Analysis of Human Response to Wildfire Events and PM_{2.5} Exposure

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Abstract

Purpose: Wildfire-induced PM_{2.5} poses an increasing public health risk due to the rising frequency and intensity of wildfires. However, its health impacts remain inadequately understood, largely because of the temporal and spatial heterogeneity of PM_{2.5} effects. This paper aims to investigate the spatiotemporal dynamics of the effect of smoke PM_{2.5} on human response and explore its interactions with environmental and socioeconomic factors.

Methods: We used Relative Research Interest (RSI) from Google Trends to assess public awareness and preparedness to wildfire smoke. We collected smoke PM_{2.5} concentration in 12 Designated Market Areas (DMAs) of California, together with the environmental and socioeconomic characteristics, from 2016 to 2020. We developed a Bayesian spatiotemporal distributed lag model to examine the temporal effect of smoke PM_{2.5} exposure on human behavior and how location-specific environmental conditions and social vulnerabilities modified the effect over time.

Results: We found 14% (95% credible interval: 12%, 17%) and 9% (95% credible interval: 7%, 12%) increase in RSI of the search topics of 'Air pollution' in the first two weeks when smoke PM_{2.5} increased by one standard deviation ($6.4 \mu\text{g}/\text{m}^3$). Similarly, we found 19% (95% credible interval: 17%, 22%) and 10% (95% credible interval: 7%, 22%) increase for search topics 'Air purifier' in the initial two weeks.

Furthermore, the increase in RSI was significantly amplified by wildfire scale, temperature, humidity, and atmospheric pressure but mitigated by wind speed. Individuals experiencing poverty or lacking health insurance were more sensitive to the increase of smoking $\text{PM}_{2.5}$, while racial/ethnic minorities and those with limited mobility or English proficiency were less sensitive.

Conclusion: Wildfire-induced $\text{PM}_{2.5}$ significantly affects human behavior, with impacts lasting for weeks. Our analysis also found significant interactions between the effect of smoke $\text{PM}_{2.5}$ and various environmental and socioeconomic factors. The disparities in people’s response to wildfire smoke, combined with inequalities in health vulnerability, underscore the importance of developing equity-focused, location-specific environmental policies to mitigate the health impacts of wildfire smoke.

Keywords: Wildfire, $\text{PM}_{2.5}$, Spatiotemporal analysis, Health equality

1 Introduction

There has been increasing interaction between wildfires and human behaviors in recent years due to climate changes, land use, and population shifts [1, 2]. Notably, the total area burned by wildfires in the western USA doubled between 1984 and 2015 [3], and the scale of wildfires is expected to escalate further as human activities increase and interactions between wildlands and urban areas grow [4, 5]. Wildfire smoke is a complex mixture of air pollutants, including particulate matter (PM) and volatile organic compounds (VOCs). Among these, $\text{PM}_{2.5}$ is recognized as the most harmful pollutant due to its high toxicity to the lungs [6, 7]. $\text{PM}_{2.5}$ is also the leading environmental risk factor for premature mortality worldwide [8], with an estimated 677,745 premature deaths attributed annually to wildfire smoke exposure [9]. Existing studies mainly rely on the daily concentration of $\text{PM}_{2.5}$ to detect the effect of $\text{PM}_{2.5}$ on human response [10–12]. However, the spatial and temporal heterogeneity of $\text{PM}_{2.5}$, as well as its correlations across different geographic locations over time, are not considered [13].

The impact of $\text{PM}_{2.5}$ varies across regions with different socioeconomic and environmental conditions. Studies have consistently shown that low-income communities, often lacking effective air filtration systems, are particularly vulnerable to smoke exposure [14–16]. Similarly, racial and ethnic minority groups report higher risks of smoke exposure, driven by systemic and structural inequalities [17, 18]. On the other hand, researchers have found that wealthier communities are more likely to seek information about wildfire smoke protection, making them better prepared to smoke $\text{PM}_{2.5}$ exposure [15, 19, 20]. These disparities in vulnerability, awareness, and preparedness regarding wildfire smoke underscore the need to study the interaction between $\text{PM}_{2.5}$ effects and socioeconomic factors. However, most existing studies focus on isolated aspects of demographic factors, leaving a gap in understanding how various factors collectively modify the effects of $\text{PM}_{2.5}$. A more comprehensive analysis that integrates multiple socioeconomic dimensions is needed to better address these disparities.

Meteorological conditions, including wind speed, humidity, atmospheric pressure, and temperature, play a critical role in determining the concentration and composition

of $\text{PM}_{2.5}$ in the atmosphere. For example, wind speed has been shown to significantly reduce $\text{PM}_{2.5}$ exposure concentrations [21]. Humidity and air pressure are associated with changes in the concentration of polycyclic aromatic hydrocarbons within $\text{PM}_{2.5}$. Additionally, higher temperatures have been linked to stronger adverse effects of $\text{PM}_{2.5}$ on mortality [16]. However, the precise influence of certain meteorological factors, such as humidity, on $\text{PM}_{2.5}$ concentrations remains poorly understood [22].

To fill these research gaps, we investigated the spatiotemporal dynamics of the effect of smoke $\text{PM}_{2.5}$ on human response in this paper. We also studied the interaction between environmental/socioeconomic factors and $\text{PM}_{2.5}$ exposure. We collected data from Google Trends (<https://trends.google.com/trends/>) for all the 12 Designated Market Areas (DMAs) in California from 2016 to 2020. Many researchers have used Google Trends as the proxy for human response to various public issues [15, 23, 24]. Google Trends measure the number of searches Google detects for specific topics, represented by the Relative Search Index (RSI). We analyzed search trends for topics of ‘Air pollution’ and ‘Air purifier’ to assess public awareness and preparedness regarding wildfire smoke (Methods). Additionally, we conducted sensitivity analyses using alternative search topics, with the results presented in the Appendix.

For environmental factors, we collected weekly meteorological data from Weather Underground (<https://www.wunderground.com/>), including temperature ($^{\circ}\text{F}$), humidity (%), atmospheric pressure (inHg), and wind speed (mph). Additionally, we gathered wildfire scale data from the California Department of Forestry and Fire Protection (<https://www.fire.ca.gov/incidents>). We measured the wildfire scale by the number of acres burned in the incident. To comprehensively assess the different aspects of socioeconomic conditions, we utilized the Social Vulnerability Index (SVI) from the Centers for Disease Control and Prevention [25]. The SVI evaluates various community aspects, such as socioeconomic status, household composition, racial and ethnic minority status, and housing type and transportation (Methods).

Utilizing the collected smoke $\text{PM}_{2.5}$, environmental, and socioeconomic data, we developed a Bayesian spatiotemporal distributed lag model by combining the Besag-York-Mollié (BYM) model [26] with a distributed lag model [27]. This model captures both spatial and temporal heterogeneity in $\text{PM}_{2.5}$ exposure. It accounts for spatial correlations in $\text{PM}_{2.5}$ concentrations across neighboring locations and incorporates temporal effects over a one-month period. We derived the metric of Relative RSI (RRSI) from the spatiotemporal distributed lag model to quantify the effect of smoke $\text{PM}_{2.5}$. RRSI represents the multiplicative change in the expected RSI for each unit increase in smoke $\text{PM}_{2.5}$ concentration (Methods).

2 Results

Fig. 1 shows the temporal trends of wildfire counts and RSI of the search terms ‘Air pollution’ and ‘Air purifier’. The trend of wildfire occurrence was highly cyclic, mostly peaking every year between June and September. RSIs for both terms showed similar but lagged cyclic pattern, peaking every year between September and December. Fig. 2 reports the temporal trends of wildfire scale measured by acres burned ($\times 10^5$ acres) and average smoke $\text{PM}_{2.5}$. The patterns were similar to that of wildfire counts, peaking

every year between June and September. Fig. 3 displays the average spatial patterns of searches for ‘Air pollution’ and ‘Air purifier’ and the spatial patterns of smoke $PM_{2.5}$ in California. The spatial pattern of RSI were different for different search terms. For ‘Air pollution’ which measures people’s awareness of wildfire smoke, we found the highest RSI in Bakersfield (6.8%) and Fresno Visalia (5.3%), while San Francisco-Oakland-San Jose (1.5%) and Sacramento-Stockton-Modesto (1.4%) showed the lowest RSI. For ‘Air purifier’ which measures people’s preparedness of wildfire smoke, San Diego (16.4%) and Los Angeles (12.3%) showed the highest RSI among 12 DMA areas, while Eureka (0.8%) and Yuma, AZ - El Centro, CA (0.4%) reported the lowest RSI. The average smoke $PM_{2.5}$ was highest in Chico Redding ($3.40 \mu g/m^3$), while it was lowest in Palm Springs ($0.31 \mu g/m^3$).

Fig. 4 shows the time-lagged effects of smoke $PM_{2.5}$ measured by relative RSI (RRSI) on Google search for ‘Air pollution’ and ‘Air purifier’. In general, we found that the increase in smoke $PM_{2.5}$ was significantly associated with the increase in Google search, and the effect of smoke $PM_{2.5}$ lasted for weeks. Specifically, one standard deviation ($6.4 \mu g/m^3$) increase in smoke $PM_{2.5}$ was associated with RRSI of 1.14 (95% credible interval: 1.12, 1.17) for ‘Air pollution’ in the same week and RRSI of 1.09 (95% credible interval: 1.07, 1.12) in the second week. For ‘Air purifier’, the RRSI was 1.19 (95% credible interval: 1.17, 1.22) in the same week, and 1.10 (95% credible interval: 1.07, 1.14) in the following week.

Fig. 5 and Fig. 6 report the cumulative RRSI for ‘Air pollution’ and ‘Air purifier’ modified by the environmental and socioeconomic factors. Among the environmental factors, fire scale, temperature, humidity, and atmospheric pressure increase the effect of smoke $PM_{2.5}$ on Google search for ‘Air pollution’ and ‘Air purifier’, while wind speed decreases the effect of $PM_{2.5}$. When fire scale increased from 1,270 acres to 32,739 acres, cumulative RRSI increased from 1.21 (95% credible interval: 1.16, 1.27) to 1.37 (95% credible interval: 1.31, 1.43) and from 1.52 (95% credible interval: 1.45, 1.60) to 1.81 (95% credible interval: 1.73, 1.90) for the search topics ‘Air pollution’ and ‘Air purifier’, respectively. We observed significant increase of cumulative RRSI for search topics of ‘Air purifier’ when humidity and atmospheric pressure increased from the 10th percentile (32.0% humidity: 1.35 (1.29, 1.42); 29.2 inHg atmospheric pressure: 1.24 (1.17, 1.33)) to the 90th percentile (82.5% humidity: 1.67 (1.55, 1.82); 29.9 inHg atmospheric pressure: 1.86 (1.74, 1.99)). We also found similar results for the search topic of ‘Air pollution’. Among the socioeconomic factors, the vulnerability in household characteristics, racial & ethnic minority status, and housing type & transportation was associated with the decrease of effect of smoke $PM_{2.5}$ on Google search for ‘Air pollution’ and ‘Air purifier’, while the vulnerability in socioeconomic status increase the effect of smoke $PM_{2.5}$. Specifically, when the vulnerability in household characteristics increased from 0.31 to 0.85, the cumulative RRSI decreased from 1.58 (95% credible interval: 1.41, 1.77) to 0.90 (95% credible interval: 0.81, 0.99) for the search topic of ‘Air pollution’ and from 2.08 (95% credible interval: 1.85, 2.35) to 1.05 (95% credible interval: 0.96, 1.16). On the other hand, when the vulnerability in socioeconomic status increased from 0.45 to 0.96, the cumulative RRSI increased from 0.96 (95% credible interval: 0.88, 1.05) to 1.75 (95% credible interval: 1.48, 2.10) for

the search topic of ‘Air pollution’ and from 1.05 (95% credible interval: 0.97, 1.14) to 2.89 (95% credible interval: 2.42, 3.43).

Fig. 7 shows the effect smoke $PM_{2.5}$ for different DMAs. We observed the largest effect of smoke $PM_{2.5}$ on ‘Air pollution’ in Eureka (RRSI: 1.68) and the smallest effect of smoke $PM_{2.5}$ in Yuma, AZ - El Centro, CA (RRSI: 0.70). Similarly, for the search topic of ‘Air purifier’, the largest effect of smoke $PM_{2.5}$ on ‘Air purifier’ was in Eureka (RRSI: 3.00), and the smallest effect of smoke $PM_{2.5}$ was in Yuma, AZ - El Centro, CA (RRSI:1.05)

3 Discussion

In this study, we found the significant effects of wildfire-induced smoke $PM_{2.5}$ on people’s awareness and preparedness of wildfire smoke. The increase in smoke $PM_{2.5}$ was associated with the higher search interest in wildfire-related topics, which was significant for lags of 0 - 1 week, but not at lag 2 or 3. Our findings are largely consistent with the existing literature. An increasing body of literature has reported the lagged effect of different outcomes. For example, Staniswalis et. al. reported the significant effect of $PM_{2.5}$ on mortality for lags of 0–2 days, but not at lag 3 [28]. Gou et. al. studied 312,917 deaths between 2015 and 2020 in Chongqing, China, and found that exposure to $PM_{2.5}$ was associated with an increased risk of mortality in the initial phase of $PM_{2.5}$ exposure (Lag 0 and Lag 1) [29]. These findings encourage policymakers to launch public awareness campaigns to educate communities about the immediate health risks of $PM_{2.5}$ exposure and emphasize protective behaviors in the early stage of the wildfire events.

Furthermore, we found the spatial heterogeneity of the effect of smoke $PM_{2.5}$ resulted from the distinctive environmental conditions in different locations. We found that fire scale, temperature, humidity, and atmospheric pressure were the positive modifiers of smoke $PM_{2.5}$, meaning that these environmental factors increased the effect of $PM_{2.5}$ on human response. The wind speed was a negative modifier of smoke $PM_{2.5}$. These findings are consistent with many studies about the meteorological effects on $PM_{2.5}$. Since wind can blow away particles, researchers have identified wind as a strong negative factor for $PM_{2.5}$ concentration [21, 30]. Research has shown that a more humid atmosphere is often associated with a lower boundary layer, which further promotes an increase in $PM_{2.5}$ concentration [31]. Higher temperatures have been found to be associated with stronger effect $PM_{2.5}$ [16]. Atmospheric pressure is positively associated with changes in the concentration of polycyclic aromatic hydrocarbons, which is a major contributor to health risk within $PM_{2.5}$ [32].

Finally, variations in socioeconomic conditions across different locations also contribute to the spatial disparities in the effects of $PM_{2.5}$. In this study, we found that individuals who were racial or ethnic minorities, lived in mobile homes, lacked transportation, or had disabilities and limited English proficiency were less sensitive to increases in smoke $PM_{2.5}$. In contrast, those experiencing poverty or lacking health insurance were more sensitive to increased exposure to smoke $PM_{2.5}$. The inequality in the effect of $PM_{2.5}$ on different health outcomes has also been reported by many

researchers. For example, Conway found that higher social vulnerability was associated with higher risks of asthma and COPD resulted from $\text{PM}_{2.5}$ [33]. Li et. al. reported that people who were socially deprived exposed to a higher level of $\text{PM}_{2.5}$ pollution [34]. These disparities in vulnerability to $\text{PM}_{2.5}$, combined with the limited awareness and preparedness observed in this study, highlight the urgent need for policymakers to develop integrated, location-specific, and justice-oriented environmental policies.

4 Limitations

The primary limitation of using Google Trends data to study the effect of $\text{PM}_{2.5}$ on human behavior is its inability to establish causation, providing only correlational insights. Google Trends reflects search behavior, an indirect measure that does not directly capture real-world actions or impacts. Additionally, the data is limited to Internet users, excluding those without access to the Internet. Selecting appropriate search terms can be challenging, potentially missing the full spectrum of relevant human behaviors. Numerous factors influence both $\text{PM}_{2.5}$ levels and human behavior, and while we have incorporated datasets to adjust for potential confounders, these adjustments may not fully account for all influencing variables. Therefore, although Google Trends can offer valuable insights into public perception and awareness of air pollution, the findings should be interpreted cautiously. The inherent limitations of the data must be acknowledged when making conclusions about the impact of $\text{PM}_{2.5}$ on human behavior.

5 Conclusion

Wildfire-induced $\text{PM}_{2.5}$ significantly affects human behavior, with impacts persisting for up to two weeks. These effects are influenced by various environmental factors, including positive modifiers such as wildfire scale, temperature, humidity, and atmospheric pressure, as well as the negative modifier of wind speed. Our research also reveals increased sensitivity to smoke $\text{PM}_{2.5}$ among individuals experiencing poverty or lacking health insurance. On the other hand, racial and ethnic minorities, as well as those with limited mobility or English proficiency, exhibit lower sensitivity. These findings underscore the importance of developing equity-focused, location-specific environmental policies to mitigate the health impacts of wildfire smoke.

6 Methods

6.1 Assessment of response to wildfire smoke

Google Trends is one of the best proxies for public interest, salience, and response to certain public issues. We accessed the data using the publicly available Trends Explore tool (<https://trends.google.com/trends/>). Google Trends reflects people’s interest in selected terms/topics, measured by the number of searches over a certain period that Google detects on Search, Google News, and YouTube. Google Trends also breaks down the data by Designated Market Areas (DMAs), the geographic areas where

Nielsen measures the local television viewing. It is discussed in [35] that Google Trends can be a valuable tool in healthcare research, providing insights into population behavior and interest over time. In this paper, we focused on the 12 DMAs in California. Google Trends comes from a random, unbiased sample of Google searches in the selected location within the study period. It is worth noticing that the data does not represent absolute search numbers, but the normalized index, Relative Search Interest (RSI), ranges from 0% to 100%, with 100% being the highest search volume and 0% being the lowest search volume.

We used Google Trends to evaluate the public responses to wildfire smoke, including awareness and preparedness. Specifically, we analyzed the search topics of ‘Air pollution’ and ‘Air purifier’. We used topics instead of search terms since topics pull in the exact phrase as well as misspellings and acronyms, and cover all languages. We collected five-year Google Trends data weekly from January 2016 to December 2020 in California. We also conducted the sensitivity analysis using Google Trends data on ‘Air quality’ and ‘Air filter’.

6.2 Assessment of wildfire smoke

The exposure to wildfire smoke was measured by smoke $PM_{2.5}$, which was the location- and time-specific anomalies of $PM_{2.5}$ on non-smoke days. The anomaly value was measured by the difference in the daily $PM_{2.5}$ and the monthly median $PM_{2.5}$. More details regarding the estimation of smoke $PM_{2.5}$ can be found in [15]. We extracted the weekly smoke $PM_{2.5}$ from January 2016 to December 2020 in California.

6.3 Assessment of effect modifiers

We collected data on the environmental and demographic factors that modified the effect of wildfire smoke on the public responses. For the environmental factors, we collected data on the wildfire scale and weather. The wildfire scale data was collected directly from the California Department of Forestry and Fire Protection (<https://www.fire.ca.gov/incidents>), including documented wildfire incidents from 2016 to 2020. We measured the wildfire scale by the number of acres burned in the incident. Meteorological data, including temperature (°F), humidity (%), atmospheric pressure (inches of mercury, inHg), and wind speed (miles per hour, mph), came from Weather Underground (<https://www.wunderground.com/>). They used a combination of personal weather stations (quality controlled), the Cooperative Observer Program from the National Weather Service, airport data, and weather balloon data.

The Social Vulnerability Index (SVI) refers to the demographic and socioeconomic factors that adversely affect communities that encounter hazards and other community-level stressors [25]. To assess the demographic status of communities thoroughly, we collected SVI describing different aspects of the communities, including the socioeconomic status, household characteristics, racial & ethnic minority status, and housing type & transportation. Specifically, socioeconomic status includes information about poverty, unemployment, housing cost burden, low education level, and the lack of health insurance. Household characteristics describe the age distribution, disability,

single parent, and English proficiency of the Household. Racial & ethnic minority status reflects minority, Black and African American, American Indian and Alaska Native, etc. Housing type & transportation show multi-structures, mobile homes, crowding, lack of vehicles, and group quarters of the communities.

6.4 Estimate the effect of wildfire smoke

The RSI at the DMA level from 2016 to 2020 was studied using spatiotemporal modeling techniques. Since smoke $PM_{2.5}$ can stay in the air for a period of time, we adopted the distributed lag models to model the effect of smoke $PM_{2.5}$ on RSI [27]. Additionally, we considered the interactions between smoke $PM_{2.5}$ and confounders mentioned above to understand how the effect of smoke $PM_{2.5}$ was modified by environmental and demographic factors. The following spatiotemporal distributed lag model was developed.

$$y_{i,t} \sim \text{Poisson}(\eta_{i,t})$$

$$\log(\eta_{i,t}) = \alpha + \sum_{l=0}^3 \left\{ (\beta_{0,l} + \sum_{j=1}^9 \beta_{j,l} \times X_{i,t-l}) \cdot PM_{2.5,i,t-l} \right\}$$

$$+ u_i + v_i + \gamma_t + \phi_t$$

where $y_{i,t}$ was the RSI for DMA i ($i = 1, 2, \dots, 12$) in California at week t ($t = 1, 2, \dots, 261$) starting from the first week in 2016. $\eta_{i,t}$ was the expected RSI. In this model, we studied the lagged effects of smoke $PM_{2.5}$ over four weeks, with $l = 0, 1, 2, 3$ weeks representing time lags. $PM_{2.5,i,t-l}$ represented the concentration of smoke $PM_{2.5}$ of DMA i at time $t-l$. The environmental conditions, including temperature, humidity, atmospheric pressure, wind speed, and fire scale of DMA i at time $t-l$, were denoted by $X_{1,t-l}, \dots, X_{5,t-l}$. $X_{6,t-l}, \dots, X_{9,t-l}$ were the socioeconomic status, household characteristics, racial & ethnic minority status, and housing type & transportation for DMA i , respectively. All variables in this model were standardized so that the mean was 0 and the standard deviation was 1.

We assumed the effect of smoke $PM_{2.5}$ lasted for four weeks in this model. Therefore, $\beta_{j,l}$'s described the l -week lagged effects of smoke $PM_{2.5}$. Specifically, $\exp(\beta_{0,l} + \sum_{j=1}^9 \beta_{j,l} X_{i,t-l})$ represented the multiplicative change in the expected RSI for a one-unit (one standard deviation) increase in the l -week's smoke $PM_{2.5}$ concentration when the standardized environmental and socioeconomic factors take the value of $X_{i,t-l}$. We defined this multiplicative change as relative RSI, i.e.,

$$\text{Relative RSI (RRSI)} = \exp(\beta_{0,l} + \sum_{j=1}^9 \beta_{j,l} X_{i,t-l})$$

where $X_{i,t-l}$ are the standardized environmental and socioeconomic factors. $\exp(\beta_{0,l})$ represents the average effect of smoke $PM_{2.5}$ fixing the modifiers at the population average. We further defined the cumulative effects of smoke $PM_{2.5}$, which is the effects

of the cumulative increase of smoke PM_{2.5} over four weeks, as follows

$$\text{Cumulative RRSI} = \exp\left(\sum_{l=0}^3 \beta_{0,l} + \sum_{l=0}^3 \sum_{j=1}^9 \beta_{j,l} X_{i,t-l}\right)$$

The u_i and v_i were the spatial random effects modeled by Besag-York-Mollier (BYM) model [26]. Under the BYM model, the spatial random effects were split into spatially-structured effect u_i and spatially-unstructured effect v_i . The conditional distribution of u_i was

$$u_i | u_{j \neq i} \sim \text{Normal}\left(\frac{1}{N_i} \sum_{j=1}^n w_{ij} u_j, \frac{\sigma_u^2}{N_i}\right)$$

where N_i was the number of neighbors for DMA i . w_{ij} was the (i, j) th element of adjacent matrix of dimension 12×12 , where $w_{ij} = 1$ if DMAs i and j were neighbors, otherwise $w_{ij} = 0$. σ_u^2 was the variance of the spatially-structured random effects. The spatially unstructured, v_i , was independently and identically normally distributed, with $v_i \sim \text{Normal}(0, \sigma_v^2)$. γ_t and ϕ_t were the temporal random effects. γ_t was the structured temporal effects modeled by a second-order random walk

$$\gamma_t | \gamma_{t-1}, \gamma_{t-2} \sim \text{Normal}(2\gamma_{t-1} - \gamma_{t-2}, \sigma_\gamma^2)$$

The temporal random effect, ϕ_t , followed independent and identical normal distribution, $\phi_i \sim \text{Normal}(0, \sigma_\phi^2)$. The spatiotemporal distributed lag model is implemented using the INLA package [36] in R version 4.3.2. Detailed implementation of the model can be found in the ‘Code availability’.

Code availability

The code to replicate all the results in the main text and supplementary is available at <https://github.com/YumingSun/spatiotemporal-wildfire>

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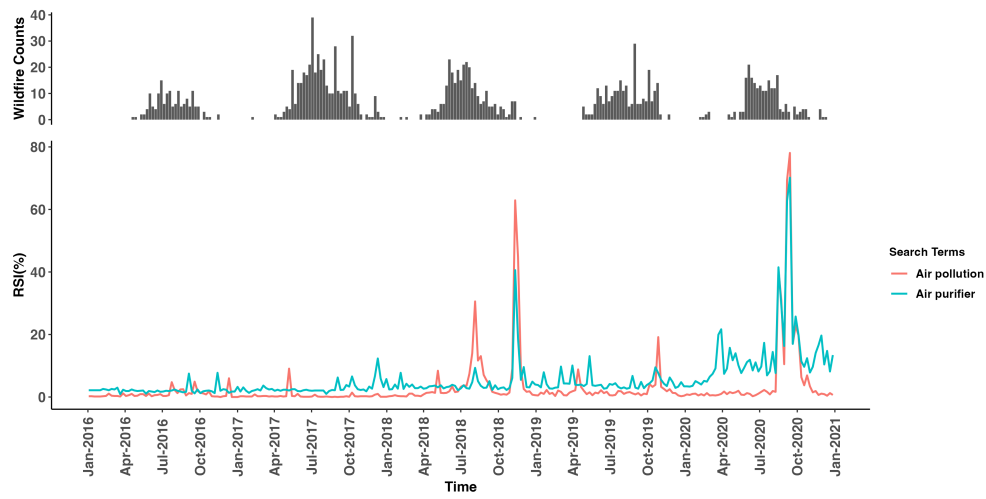


Fig. 1 Temporal trends of wildfire counts and public interest in wildfire from 2016 to 2020. The upper panel shows the temporal trends of average wildfire counts of 12 Designated Market Areas (DMAs) in California. The lower panel shows the temporal trends of the average Relative Search Interest (RSI,%) of 12 DMAs in California.

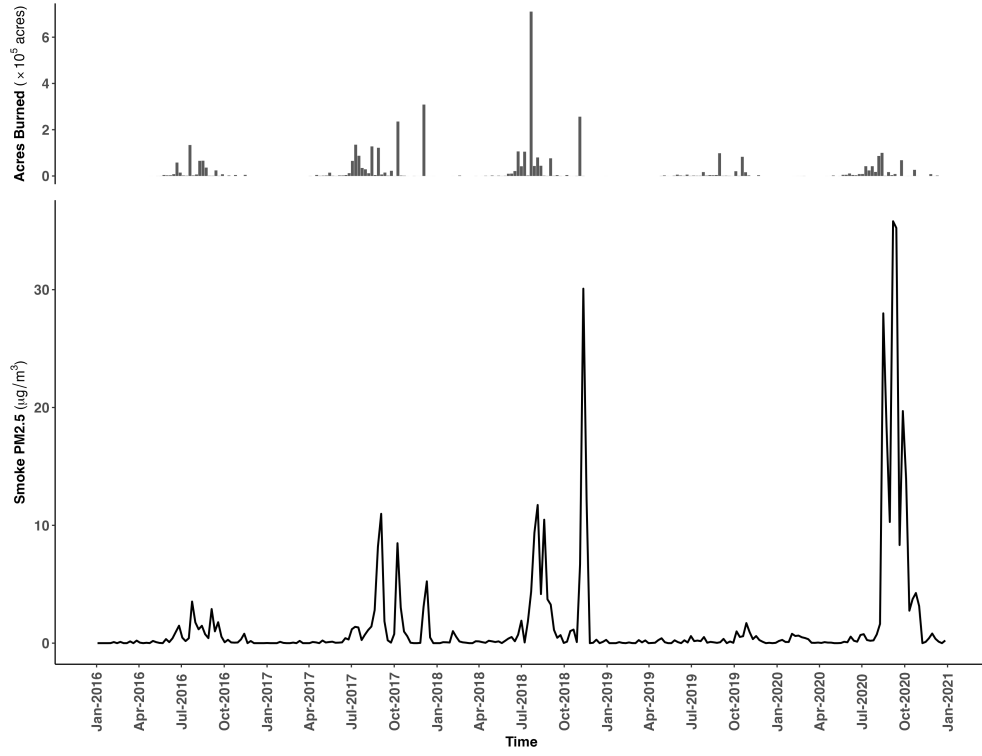


Fig. 2 Temporal trends of wildfire scale and smoke PM_{2.5} from 2016 to 2020. The upper panel shows the temporal trends of the average acres burned ($\times 10^5$ acres) in the 12 Designated Market Areas (DMAs) in California. The lower panel shows the temporal trends of average smoke PM_{2.5} ($\mu\text{g}/\text{m}^3$) of 12 DMAs in California.

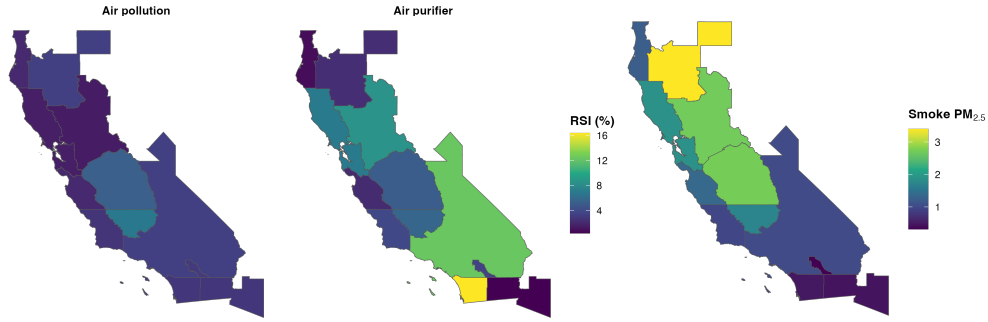


Fig. 3 Spatial trends of public interest and smoke $\text{PM}_{2.5}$ in 12 Designated Market Areas (DMAs) in California. The two left panels show spatial trends of the average Relative Search Interest (RSI, %) from 2016 to 2020. The left panel shows the spatial trends of the average smoke $\text{PM}_{2.5}$ ($\mu\text{g}/\text{m}^3$) from 2016 to 2020.

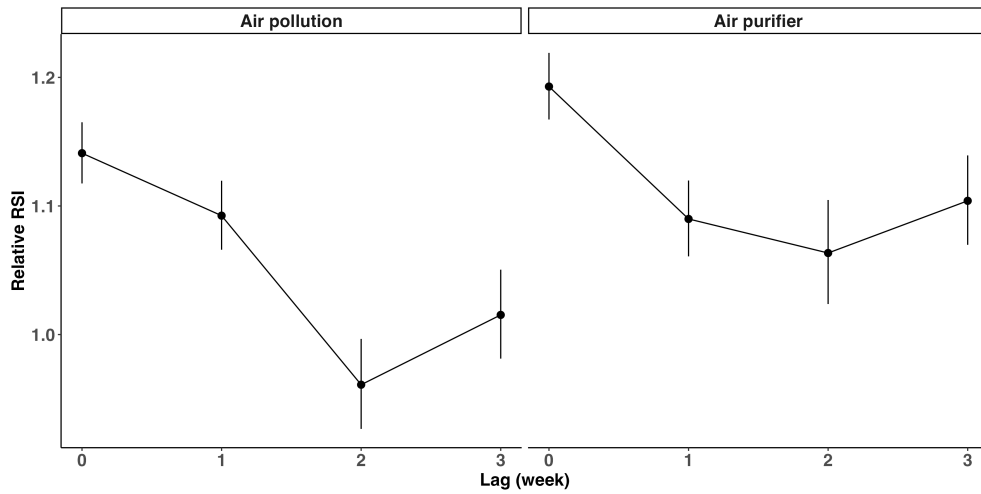


Fig. 4 Average time-lagged effect of smoke $\text{PM}_{2.5}$ ($\mu\text{g}/\text{m}^3$) measured by Relative RSI. The average relative RSI is calculated fixing other environmental and socioeconomic factors at the population average.

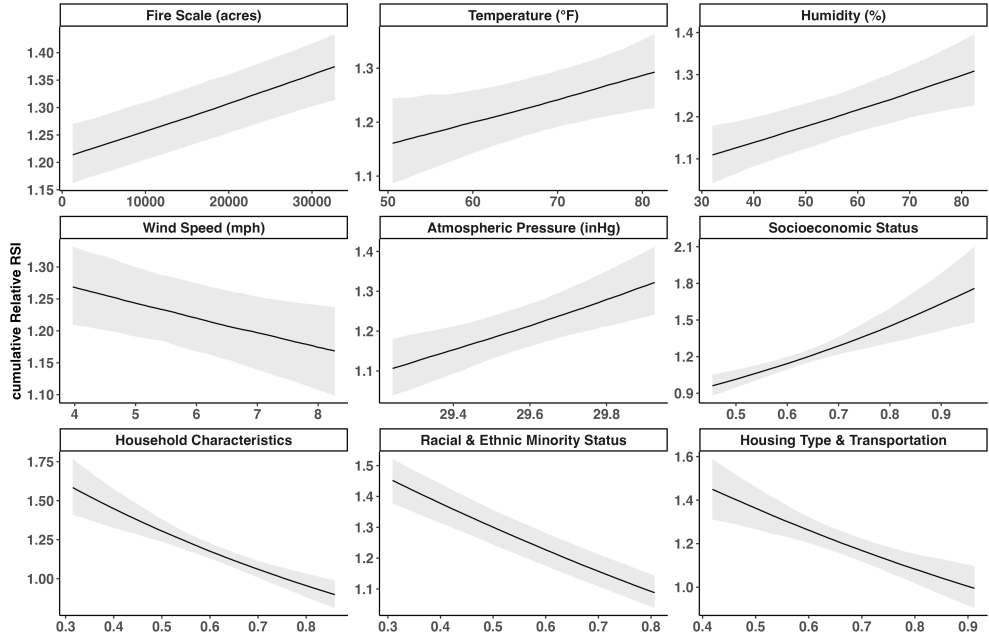


Fig. 5 Cumulative effect of smoke $PM_{2.5}$ ($\mu g/m^3$) on Google search for 'Air pollution' modified by environmental and socioeconomic factors fixing other modifiers at the population average.

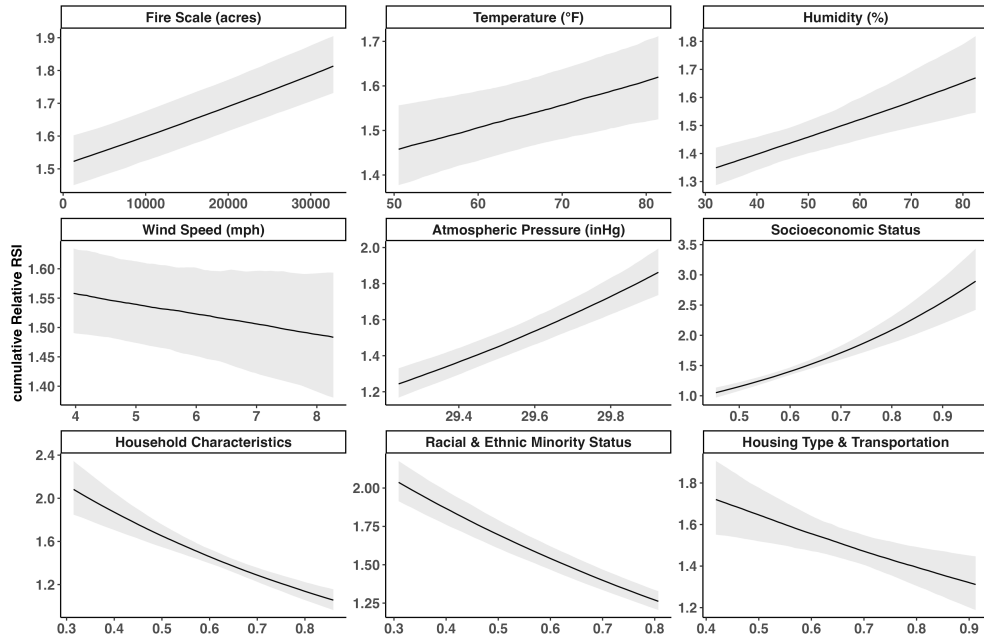


Fig. 6 Cumulative effect of smoke $PM_{2.5}$ ($\mu g/m^3$) on Google search for 'Air purifier' modified by environmental and socioeconomic factors fixing other modifiers at the population average.

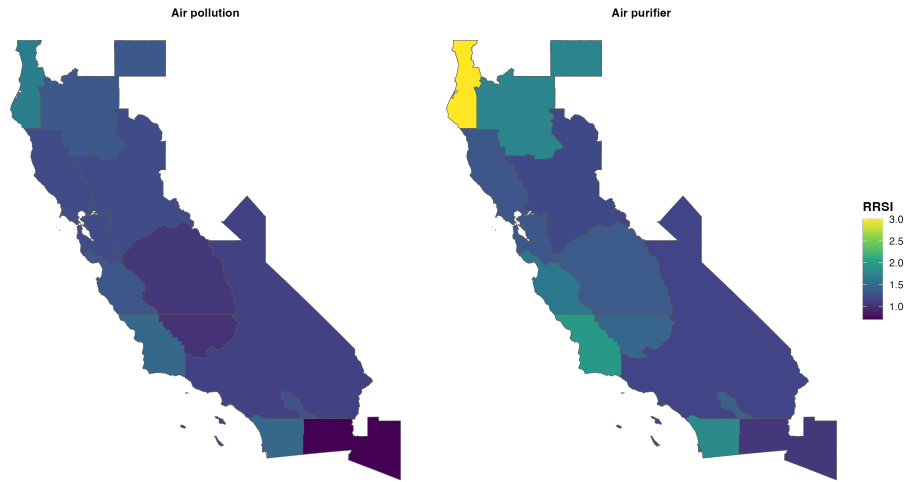


Fig. 7 Effect of smoke $PM_{2.5}$ ($\mu g/m^3$) measured by relative RSI (RRSI) for different Designated Market Areas (DMAs) in California.

Appendix A Sensitivity Analysis

We conducted the sensitivity analysis using alternative air-pollution-related search terms, including ‘Air quality’ and ‘Air filter’.

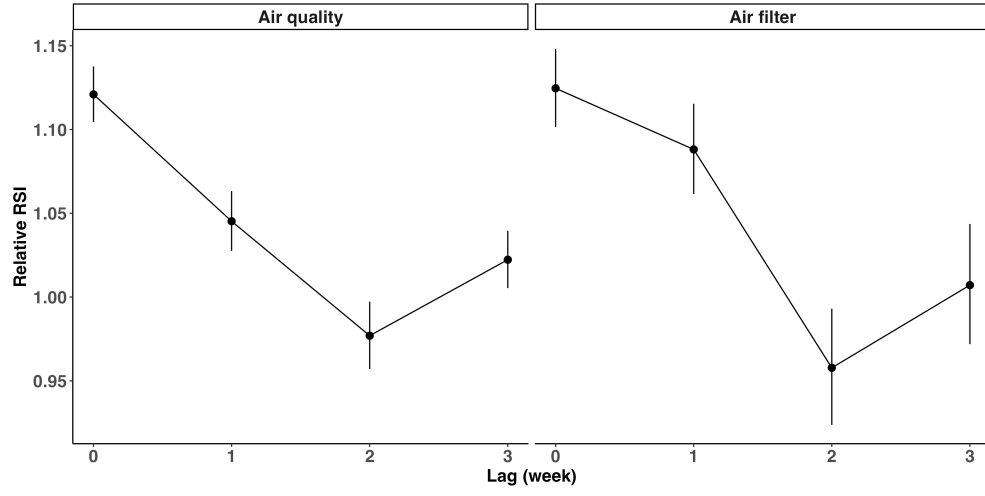


Fig. A1 Average time-lagged effect of smoke $\text{PM}_{2.5}$ ($\mu\text{g}/\text{m}^3$) on Google search terms of ‘Air quality’ and ‘Air pollution’.

Table A1 Estimated coefficients (95% credible interval) of the environmental and demographic modifiers for the effect of PM_{2.5} on Google search for ‘Air quality’¹

Modifiers	Lag0	Lag1	Lag2	Lag3
Fire Scale	0.025 (0.021, 0.030)	0.031 (0.027, 0.035)	0.018 (0.014, 0.023)	0.013 (0.007, 0.018)
Temperature	-0.006 (-0.020, 0.007)	0.015 (-0.001, 0.033)	0.029 (0.002, 0.054)	-0.054 (-0.084, -0.026)
Humidity	0.008 (-0.009, 0.027)	0.020 (-0.001, 0.042)	-0.009 (-0.038, 0.022)	-0.016 (-0.045, 0.014)
Wind Speed	0.003 (-0.007, 0.014)	-0.001 (-0.016, 0.014)	-0.041 (-0.064, -0.020)	-0.017 (-0.040, 0.008)
Atmospheric Pressure	-0.029 (-0.048, -0.008)	-0.011 (-0.035, 0.014)	0.065 (0.027, 0.103)	0.056 (0.022, 0.091)
Socioeconomic Status	-0.002 (-0.054, 0.053)	0.024 (-0.037, 0.086)	-0.030 (-0.121, 0.051)	0.162 (0.081, 0.249)
Household Characteristics	-0.076 (-0.119, -0.034)	-0.043 (-0.094, 0.008)	0.034 (-0.031, 0.107)	-0.063 (-0.130, -0.001)
Racial & Ethnic Minority	-0.004 (-0.020, 0.012)	-0.018 (-0.036, -0.001)	-0.048 (-0.071, -0.025)	-0.037 (-0.056, -0.018)
Housing Type & Transportation	0.007 (-0.027, 0.039)	-0.011 (-0.051, 0.031)	-0.002 (-0.058, 0.052)	-0.108 (-0.161, -0.055)

¹Coefficients from the distributed lag models at different time lags are reported.

Table A2 Estimated coefficients (95% credible interval) of the environmental and demographic modifiers for the effect of PM_{2.5} on Google search for ‘Air filter’¹

Modifiers	Lag0	Lag1	Lag2	Lag3
Fire Scale	0.006 (0.002, 0.009)	0.008 (0.005, 0.011)	0.008 (0.004, 0.011)	0.007 (0.004, 0.011)
Temperature	-0.023 (-0.033, -0.013)	-0.013 (-0.024, -0.002)	-0.011 (-0.022, 0.002)	-0.022 (-0.033, -0.011)
Humidity	0.030 (0.017, 0.043)	0.019 (0.003, 0.035)	-0.029 (-0.047, -0.012)	0.012 (-0.003, 0.027)
Wind Speed	0.005 (-0.004, 0.013)	-0.005 (-0.015, 0.005)	0.020 (0.009, 0.030)	-0.002 (-0.012, 0.009)
Atmospheric Pressure	0.008 (-0.007, 0.024)	0.002 (-0.014, 0.017)	0.006 (-0.010, 0.024)	0.035 (0.019, 0.051)
Socioeconomic Status	-0.029 (-0.071, 0.011)	0.019 (-0.024, 0.062)	-0.021 (-0.070, 0.026)	0.139 (0.092, 0.187)
Household Characteristics	0.023 (-0.010, 0.058)	-0.014 (-0.051, 0.021)	0.012 (-0.026, 0.051)	-0.067 (-0.106, -0.032)
Racial & Ethnic Minority	-0.006 (-0.019, 0.006)	-0.045 (-0.058, -0.032)	0.052 (0.034, 0.068)	-0.048 (-0.061, -0.037)
Housing Type & Transportation	0.021 (-0.006, 0.049)	0.023 (-0.004, 0.053)	-0.024 (-0.056, 0.007)	-0.077 (-0.109, -0.043)

¹Coefficients from the distributed lag models at different time lags are reported.



Fig. A2 Effect of smoke PM_{2.5} ($\mu\text{g}/\text{m}^3$) measured by relative RSI (RRSI) for different Designated Market Areas (DMAs) in California.